Appendix C: IEEE 830 Template

Software Requirements Specification

for

E-commerce Application

Version 1.0

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Revision History

Name	Date	Reason for Changes	Version
Jorge Domínguez	07/04/2026	Initial version	1.0
Jorge Dominguez	11/04/2026	Added UML diagrams (Class, Sequence, Activity)	1.1
Jorge Dominguez	19/04/2026	Completed system description	1.2

1. Introduction

1.1. Purpose

This document describes the requirements for an E-commerce Application that allows users to browse products, make purchases, and manage orders.

1.2. Document Conventions

This document is intended for developers, testers, and project managers involved in the development of the system.

1.3. Intended Audience and Reading Suggestions

This document is intended for developers, testers, project managers, and end users.

Developers will use it to implement the system, testers to validate requirements, and users to understand system capabilities.

1.4. Product Scope

The system will allow users to browse products, add items to a cart, and complete purchases. It aims to provide a simple and efficient online shopping experience.

1.5. References

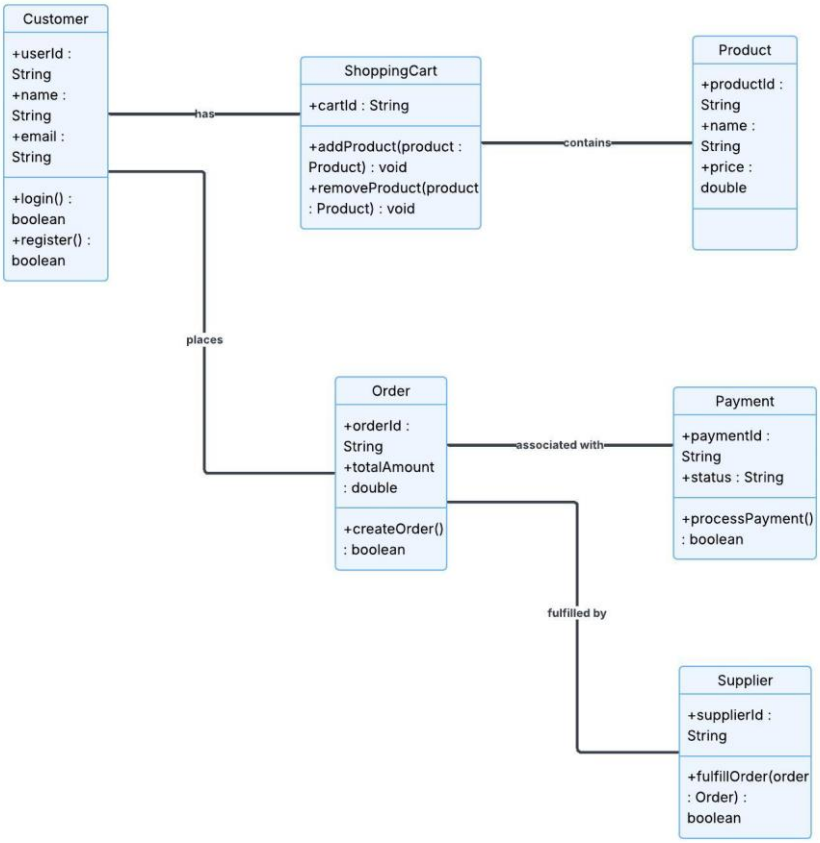
- Software Engineering Lab Materials (Lodz University of Technology)

- IEEE 830 Software Requirements Specification Standard

2. Overall Description

2.1. Product Perspective

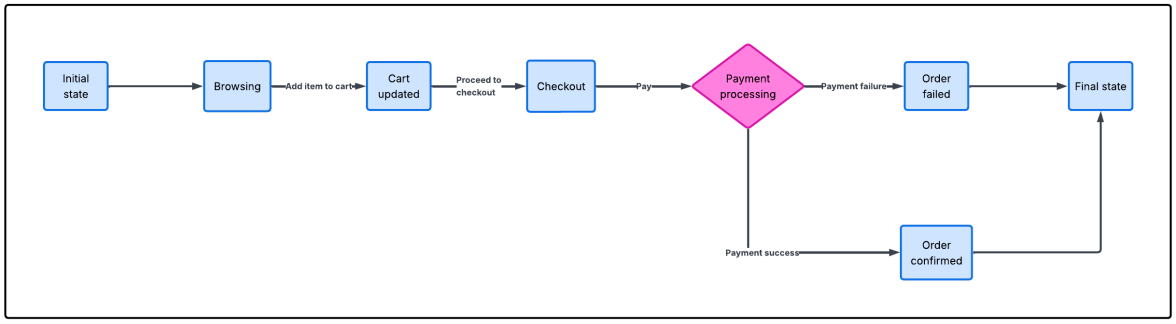
The following class diagram presents the high-level structural model of the e-commerce system. It shows the main entities involved in the application, including customers, products, shopping carts, orders, payments, and suppliers, together with their relationships.



2.1.1 State Diagram

The following state diagram represents the different states of an order in the e-commerce system. It shows how the order transitions from browsing and cart updates to checkout, payment processing, and final states depending on the success or failure of the payment

This diagram provides a high-level view of the system behavior during the order lifecycle.



2.2. Product Features

- User registration and login
- Product browsing
- Add products to cart
- Checkout and payment
- Order management

2.3. User Classes and Characteristics

- Customer (regular users)
- Admin (manages products and orders)

2.4. Operating Environment

The system will operate as a web-based application accessible through modern web browsers. It is designed to run on different operating systems, including Windows, macOS, and Linux, as well as mobile platforms such as Android and iOS.

Users will access the system using widely adopted browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The application will rely on a server-side environment capable of handling web requests and managing persistent data through a database system.

Additionally, the system will interact with external services, such as payment gateways and supplier systems, through API-based communication, ensuring seamless integration with third-party components.

2.5. Design and Implementation Constraints

The system must be developed as a web-based application using standard web technologies, ensuring compatibility with modern browsers and devices. It is constrained by the need to integrate with external payment services, which are responsible for securely processing financial transactions.

Furthermore, the system depends on supplier platforms for order fulfillment, which introduces limitations related to external system availability and communication protocols. Security considerations also impose constraints, requiring the system to protect user data and handle payment information in a secure manner.

The application must support multiple users simultaneously and follow standard design and development conventions to ensure maintainability, scalability, and usability.

2.6. User Documentation

The system will include user documentation to support users in understanding and interacting with the application. This documentation will provide guidance on how to browse products, manage the shopping cart, and complete purchases.

In addition to formal documentation, the system will offer built-in assistance through clear interface elements, informative feedback messages, and error notifications. These elements will help users navigate the system and resolve common issues during interaction.

All documentation will be provided in digital format and will be accessible directly within the application

.2.7. Assumptions and Dependencies

The system assumes that users have access to a stable internet connection and use modern web browsers that support current web standards. It also relies on the availability and correct functioning of external services, such as payment gateways and supplier systems, which are essential for processing transactions and fulfilling orders.

The application depends on a database system to store and manage user information, product data, and order records. Any disruption in these external components or services may affect the overall functionality of the system

.3. System Features

3.1. User Management

Allows users to register and log into the system.

3.2. Product Management

Allows browsing and viewing product details.

3.3. Shopping Cart

Allows users to add and remove products.

3.4. Order Processing

Allows users to place orders and complete payments

4. External Interface Requirements

4.1. User Interfaces Overview

The system will provide a graphical user interface (GUI) accessible through modern web browsers. Users will interact with the system using standard web elements such as buttons, forms, and navigation menus.

From the user's perspective, the system will allow:

- Browsing products through a product catalog page

- Viewing detailed information for each product

- Adding and removing items from the shopping cart

- Proceeding to checkout and completing payment

- Viewing order confirmation and order history

The interface will provide immediate feedback for user actions, such as confirmation messages when items are added to the cart or error messages if payment fails.

The interface will follow standard usability principles, including:

- Clear navigation menus

- Consistent layout across all pages

- Responsive design for different screen sizes (desktop and mobile)

Standard GUI elements will include:

- Buttons for actions such as "Add to Cart", "Checkout", and "Pay"

- Input fields for user data (email, password, payment details)

- Notifications for success or failure of operations

Error messages will be displayed clearly to inform users of issues such as invalid login credentials or payment failures.

4.2. Hardware Interfaces

The system does not require direct interaction with specific hardware components. It will operate as a web-based application accessible from standard user devices.

Supported devices include:

- Desktop computers

- Laptops

- Tablets

- Smartphones

The system will rely on standard hardware components such as:

- Input devices (keyboard, mouse, touch screen)

- Display devices (screens)

No specialized hardware interfaces are required. All interactions occur through the user's web browser.

4.3. Software Interfaces

The system interacts with several external software components to provide its functionality.

- Payment Gateway:

  The system integrates with a third-party payment service to process transactions securely. The payment system handles payment authorization, validation, and confirmation.

- Supplier System:

  The system communicates with external supplier systems to fulfill customer orders. Once an order is confirmed, the order details are sent to the supplier for processing and shipment.

- Database System:

  The application uses a database to store user data, product information, shopping cart contents, and order records.

- Web Browser:

  The system is accessed through modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.

Data exchange between the system and external components is performed through APIs. These interfaces ensure secure and reliable communication between the e-commerce platform and external services

.5. System Features/Modules

5.1. User Management

5.1.1 Description and Priority

This feature allows users to register and log into the system.
Priority: High

5.1.2 Stimulus/Response Sequences

Stimulus: The user enters email and password
Response: The system authenticates the user and grants access

5.1.3 Functional Requirements

some action>." Note that the entire requirement must be stated in the requirements area. Everything else is just a lead-up to this. There should be one such entry per S/R pair.»

REQ-1.1: The system shall allow users to register using email and password.
REQ-1.2: The system shall allow users to log in securely.

5.2. Product Browsing

5.2.1 Description and Priority

This feature allows users to view and search products.
Priority: High

5.2.2 Stimulus/Response Sequences

Stimulus: The user selects a product
Response: The system displays product details

5.2.3 Functional Requirements

REQ-2.1: The system shall display a list of products.
REQ-2.2: The system shall show detailed product information.

5.3. Shopping Cart

5.3.1 Description and Priority

This feature allows users to manage their shopping cart.
Priority: High

5.3.2 Stimulus/Response Sequences

Stimulus: The user adds a product to the cart
Response: The system updates the cart

5.3.3 Functional Requirements

REQ-3.1: The system shall allow users to add products to the cart.
REQ-3.2: The system shall allow users to remove products from the cart.

5.4. Order Processing

5.4.1 Description and Priority

This feature allows users to complete purchases.
Priority: High

5.4.2 Stimulus/Response Sequences

Stimulus: The user confirms the order
Response: The system processes the payment and creates the order

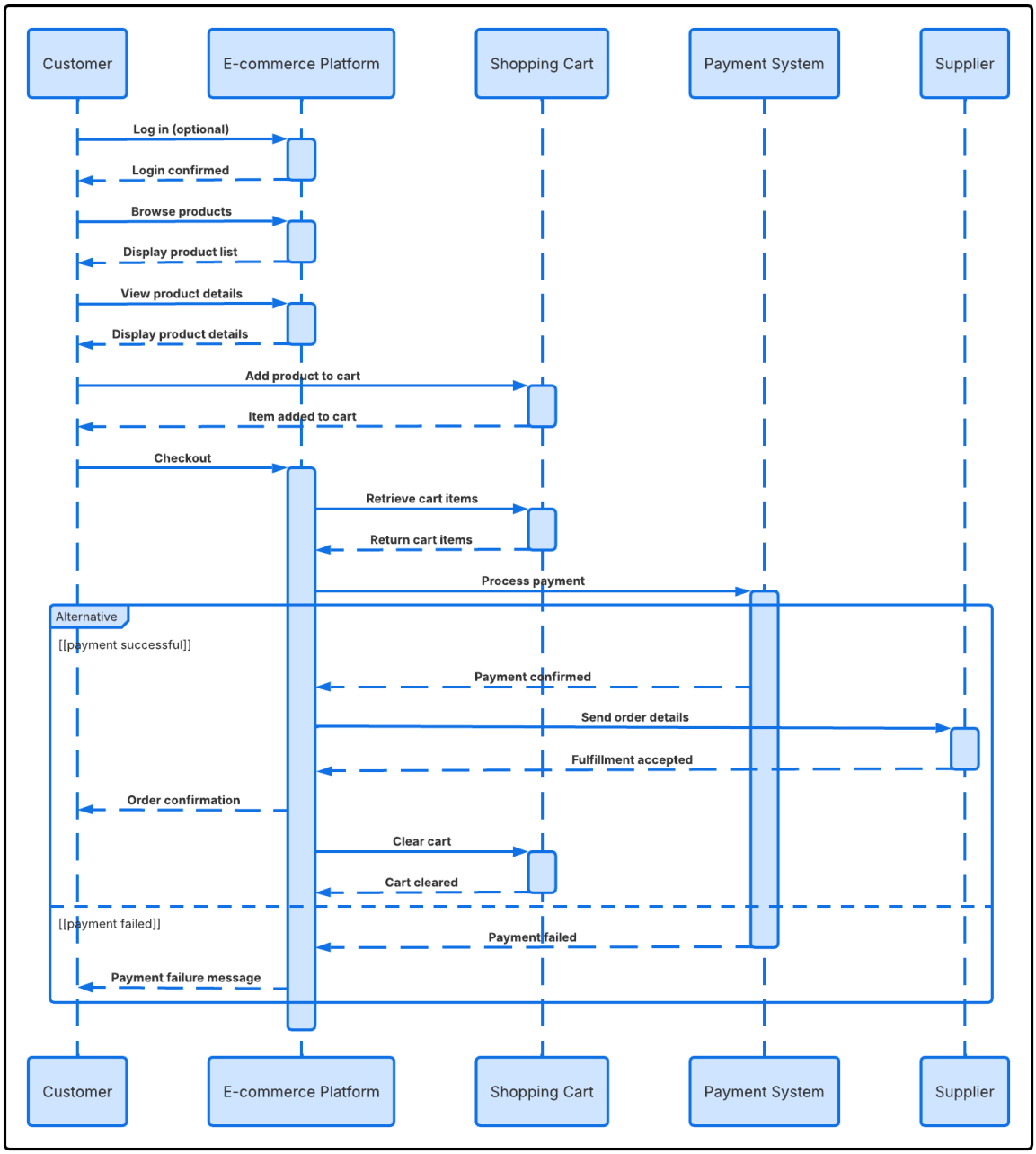
5.4.3 Functional Requirements

REQ-4.1: The system shall allow users to place orders.
REQ-4.2: The system shall process payments securely.

5.4.4 Sequence Diagram

The following sequence diagram illustrates the order processing workflow in the system. It shows how the customer interacts with the e-commerce platform to browse products, add items to the cart, proceed to checkout, and complete the payment process.

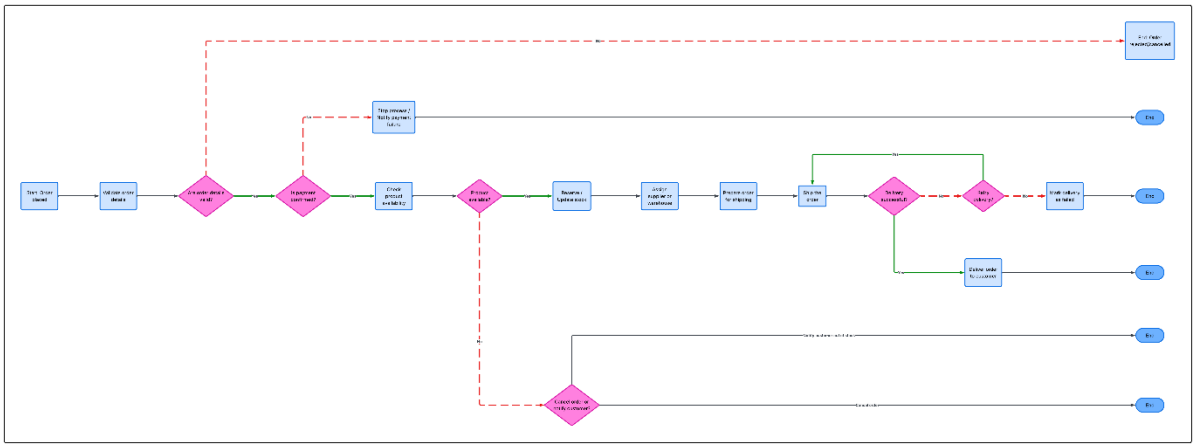
The diagram also includes interactions with external systems such as the payment system and supplier, as well as alternative flows for successful and failed payments.



5.4.5 Activity Diagram

The following activity diagram illustrates the internal workflow of the order processing system. It shows the different steps involved after an order is placed, including validation, payment confirmation, stock management, shipping, and delivery.

The diagram also includes alternative flows such as payment failure, unavailable products, and delivery issues.



6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The system shall respond to user actions in less than 2 seconds.

6.2. Security

NF-2.1: The system shall encrypt user data and payment information.

6.3 Reliability

NF-3.1: The system shall maintain at least 99% uptime.